Accessibility selects the circuit: a controlled 2+2→4 vs. 1+1+1+1→4 experiment

4 relevant bits, full 2^4 cube, two ReLU residual blocks, no weight decay — exact Walsh measurements

Question and prediction

When a fresh residual block learns the degree-4 parity $x_1x_2x_3x_4$, does it combine the four *singletons* ($1^4 \to 4$) or the two already-built degree-2 *parents* ($2{+}2 \to 4$)? For a balanced p-parent ReLU gate, $\gamma = |v||q|/\sqrt{p}$ and the produced target coefficient is $c_p = \kappa_p \gamma$, so the production per unit hidden weight is $|c_p|/|v| = |\kappa_p|\,|q|/\sqrt{p}$: with $|\kappa_4| = \frac{1}{4}$ the singleton route yields $Q_1/8$, with $|\kappa_2| = \frac{1}{2}$ the parent route yields $Q_2/2\sqrt{2}$. The compositional route should win exactly when the *measured accessibility ratio* crosses

$$R = \frac{Q_2}{Q_1} > R_c = \frac{1}{\sqrt{8}} \approx 0.354, \qquad Q_1 = \|q_{\{i\}}\|, \quad Q_2 = \|q_{\{i,j\}}\|, \quad q_S = \mathbb{E}_x[h(x)\chi_S(x)]\,.$$

Protocol

Residual network $h^{(\ell+1)} = h^{(\ell)} + \mathrm{ReLU}(V^{(\ell)}h^{(\ell)})$, width $N = 1024$, all weights (embedding, blocks, readout) initialized with variance $1/N$, $\lambda_{\mathrm{wd}} = 0$, full-batch SGD on all $2^4 = 16$ inputs (exact expectations, float64; lr 0.01, momentum 0.9, convergence-based stopping). **Stage 1:** train embedding + block 1 on the parents x_1x_2, x_3x_4 to machine precision; the pair modes are produced by textbook $1{+}1 \to 2$ corner units (contribution-weighted singleton share of their pre-activations = 1.000 in every seed). Freeze. **Rescaling:** exploit ReLU homogeneity, $h_1(\alpha) = h_0 + \alpha\,\mathrm{ReLU}(V_1h_0)$, and *re-measure* $Q_1(\alpha), Q_2(\alpha)$ directly — $\alpha \in [0, 8]$ maps to measured $R \in [0, 0.79]$. **Stage 2:** for each α (13 values × 4 seeds), train an *identically initialized* fresh block 2 + readout on $x_1x_2x_3x_4$. Every run reaches accuracy 1.0 at loss $\sim 10^{-28}$ — the task is always solved; only the *mechanism* changes.

Classification is by pre-activation, not activation (both routes emit the same χ_{1234}): for the block-2 neurons covering 95% of $\sum_j |w_j c_{4,j}|$ ($c_{4,j} = \mathbb{E}[\mathrm{ReLU}(z_j)\chi_{1234}]$), decompose $\mathrm{Var}\, z_j$ exactly over the cube and take the share on the two parents vs. the four singletons; the reported fraction is the $|w_j c_{4,j}|$-weighted average of these shares.

Result

The transition exists and sits at the predicted boundary (Fig. 1): for all $R \le 0.27$ the parent share is ≤ 0.03 — block 2 is a pure singleton machine — and the first substantial 2+2 fraction appears at $R \approx 0.38$, immediately past R_c (0.10–0.14), growing to ≈ 0.28 at $R \approx 0.47$, ≈ 0.45 at $R \approx 0.6$–0.73, and 0.61–0.71 at $R \approx 0.77$; the singleton panel mirrors it. Seeds collapse onto a single curve: the measured R, not the seed, determines the mix. The onset is a soft crossover starting at R_c (parity near $R \approx 0.7$), as expected when two routes of near-equal efficiency are populated by gradient descent. *Caveat:* block-1 corner units leak a small $d4$ into the frozen stream ($\|q_{1234}\| \approx 0.105\,\alpha$); by $\alpha = 8$ the readout solves the task without block 2 (no contributing neurons — excluded), so the rightmost points describe only what block 2 adds.

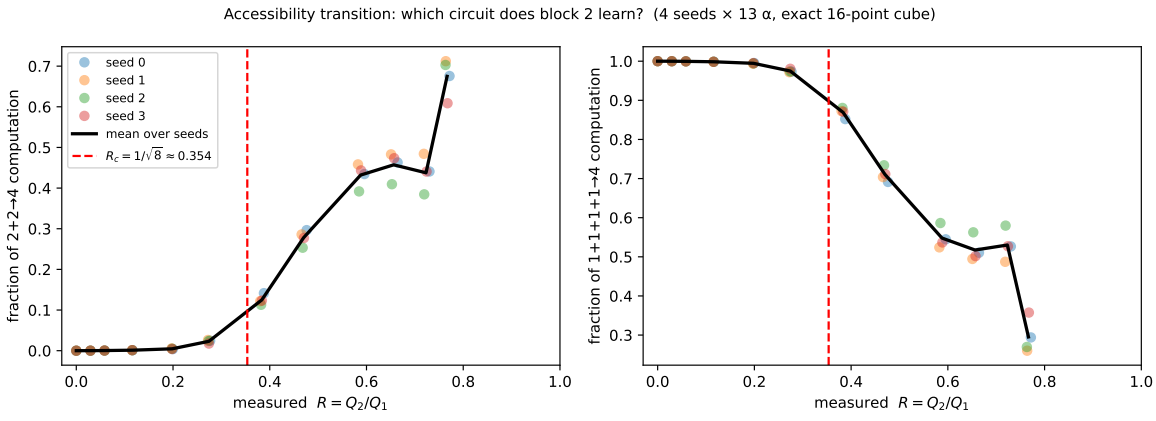


Figure 1: Fraction of $2{+}2 \to 4$ (left) and $1^4 \to 4$ (right) computation in block 2 vs. the measured accessibility ratio $R = Q_2/Q_1$ of the frozen representation; red dashed: $R_c = 1/\sqrt{8}$. Points: 4 seeds × 13 α; black: mean over seeds.

Artifacts: `gdrive:parity/analysis_output/accessibility_4bit/` (`results.csv`, `trajectories.csv` — classification is stable well before stopping — `block2_neurons.csv`, `stage1.json`, figures, this report); code: `analysis/accessibility/run_experiment.py`.